
# CHEM F111 : General Chemistry

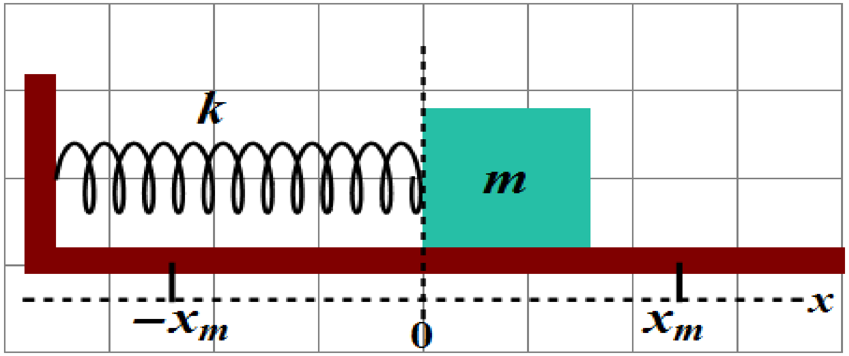
## Semester II: AY 2023-24

Lecture- 05, 24th January 2024, Wednesday innovate achieve lead

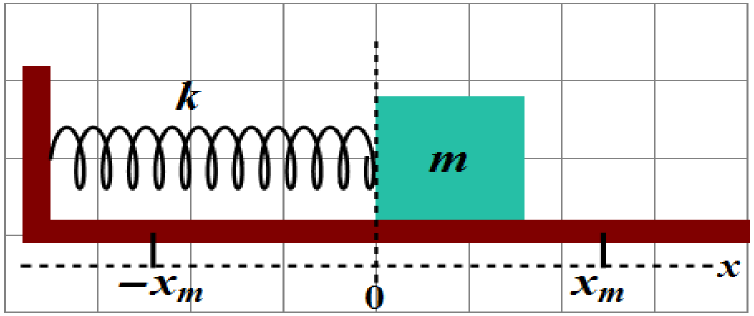
## Last Class:

- Particle in a one-dimensional box; normalized wavefunction, energy, zero-point energy, quantum numbers (due to boundary conditions), transitions;
- Symmetric and anti-symmetric nature of the wavefunction of PIB; nodes;
- Correspondence principle;
- Particle in a 2D- potential energy free box; visualization of wave functions; Particle in a 3D- potential energy free box.

# Example 2: Application of Quantum Mechanics to Vibrational motion



# Vibrational motion: classical mechanics



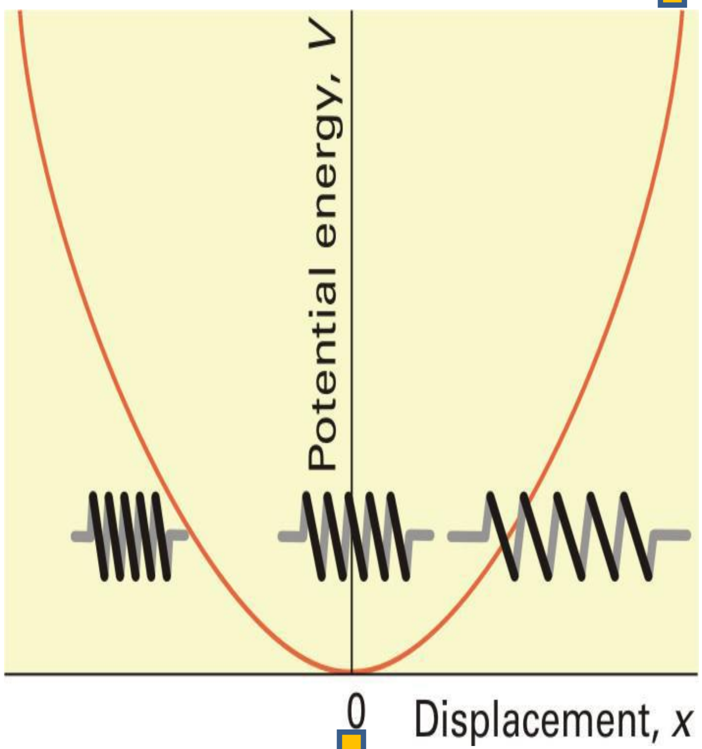

**Turning point; K.E = 0; V= max.**

One-Dimensional displacement x from equilibrium position

Force acting against the displacement x; Hookes's Law: $F = -kx$

$$V(x) = \int -F dx = \int -(-kx) dx = \frac{kx^2}{2}$$

$x$ can take on any value. Energy is continuous, continuous range of values.

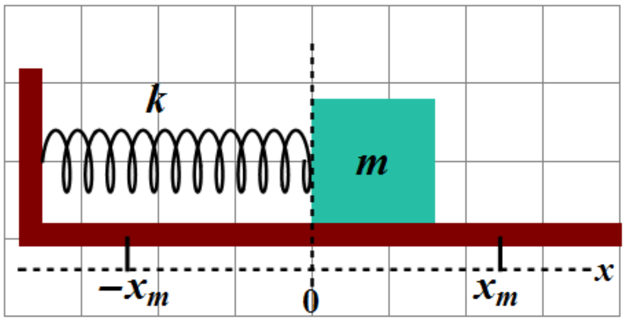



***x* = 0; *V* = min. (0); K.E = max.**

**Parabolic Potential Energy**

# Vibrational motion: quantum mechanics

A wavefunction for a one-dimensional harmonic oscillator can be obtained by solving the time-independent Schrödinger equation:




$$\hat{H}\psi = E\psi$$

$$\hat{H} = -\frac{\hbar^2}{2m}\frac{d^2}{dx^2} + \frac{1}{2}kx^2$$

**kinetic energy** **potential energy**

$$\left[-\frac{\hbar^2}{2m}\frac{d^2}{dx^2} + \frac{1}{2}kx^2\right]\psi = E\psi$$

***(derivation not in syllabus)***

Solve the Schrodinger equation and apply the boundary conditions (Boundary conditions: $\psi \rightarrow 0$ as $x \rightarrow \pm\infty$):

# QM treatment of harmonic oscillator: the solutions

## Wavefunction

$$\psi_n = N_n H_n(y)\ e^{-\alpha x^2/2}$$

$$y = x\sqrt{\alpha}$$

$N_n$: *Normalization constant*

$$N_n = \frac{1}{\sqrt{2^n n!}}\left(\frac{\alpha}{\pi}\right)^{1/4}$$

$H_n$: *Hermite polynomials*

$$H_n(y) = (-1)^n e^{y^2}\frac{d^n}{dy^n}e^{-y^2}$$

$n$: *Vibrational quantum No.*

$$n = 0,1,2,\ldots$$

$$\alpha = \frac{4\pi^2 \nu m}{h}$$

***Vibrational frequency***

$$\nu = \frac{1}{2\pi}\left(\frac{k}{m}\right)^{1/2}$$

$$\psi_0 = \left(\frac{\alpha}{\pi}\right)^{\frac{1}{4}}.e^{-\alpha x^2/2}$$

$$\psi_1 = \left(\frac{\alpha}{\pi}\right)^{\frac{1}{4}}.\left(\sqrt{2\alpha}.x\right).e^{-\alpha x^2/2}$$

$$\psi_2 = \left(\frac{\alpha}{\pi}\right)^{\frac{1}{4}}.\left(\frac{1}{\sqrt{2}}\left(2\alpha x^2 - 1\right)\right).e^{-\alpha x^2/2}$$

Energy of the oscillator:

$$E_n = \left(n + \frac{1}{2}\right)h\nu\ ; \quad n = 0,1,2,\ldots$$

Vibrational quantum number: n

# 1D harmonic oscillator: Hermite polynomials

Hermite polynomials can be easily obtained using the following formula:

$$H_n(y) = (-1)^n e^{y^2} \frac{d^n}{dy^n} e^{-y^2} \qquad n = 0,1,2,\ldots$$

$$H_0(y)=1$$
$$H_1(y)=2y$$
$$H_2(y)=4y^2-2$$
$$H_3(y)=8y^3-12y$$
$$H_4(y)=16y^4-48y^2+12$$
$$H_5(y)=32y^5-160y^3+120y$$

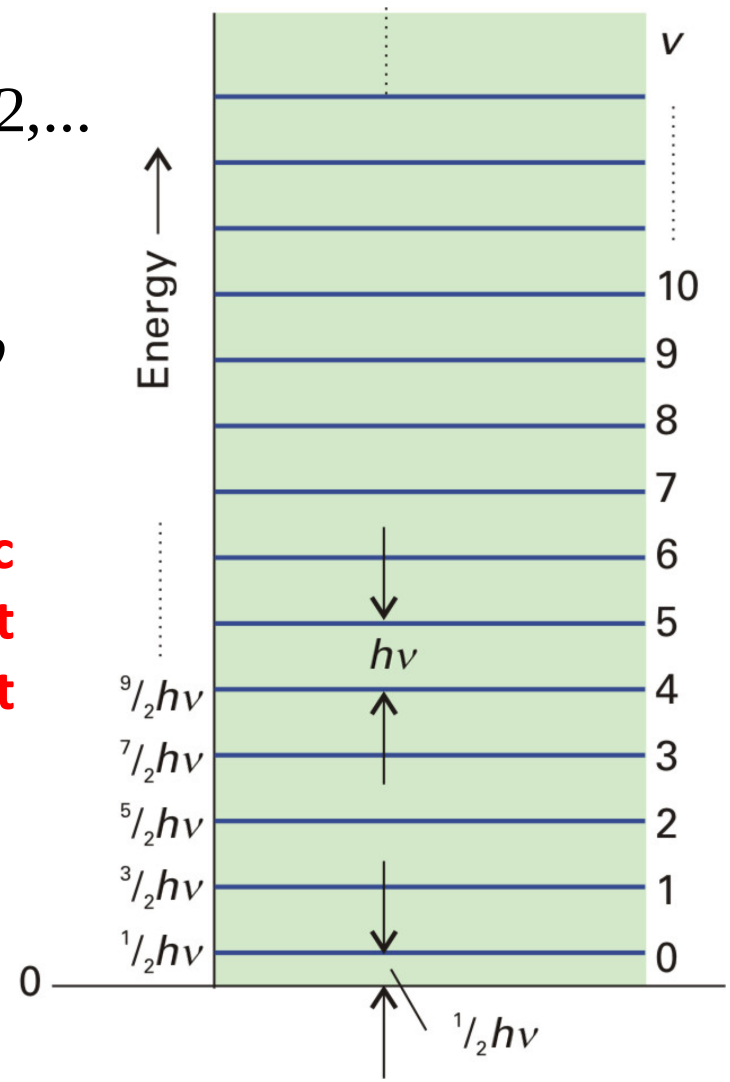
$$y = x\sqrt{\alpha}$$

$$\psi_v(y)=N_v H_v(y) e^{-y^2/2}$$

# QM harmonic oscillator: total energy eigen value

**Energy of the oscillator:**

$$E_n = \left(n + \frac{1}{2}\right) h\nu \,; \quad n = 0,1,2,\ldots$$




**Ground state energy:** $E_0 = \frac{1}{2} h\nu = \frac{1}{2}\hbar\omega$

**(Zero-point energy: *n* = 0)**

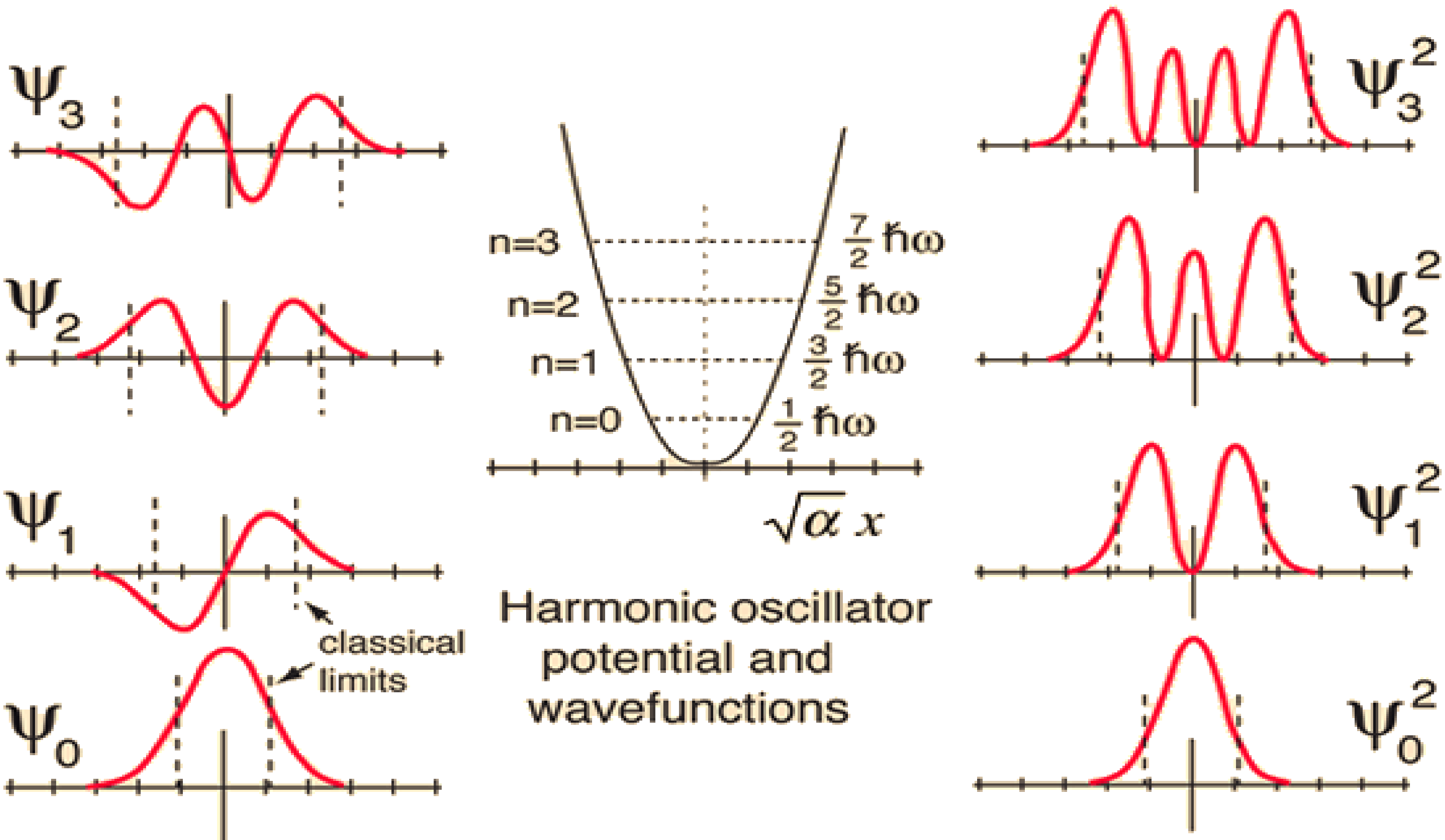
**Atoms in a solid lattice or in polyatomic molecules cannot have zero energy even at absolute zero temperature. "zero-point vibration".**

**Evenly spaced energy levels.**

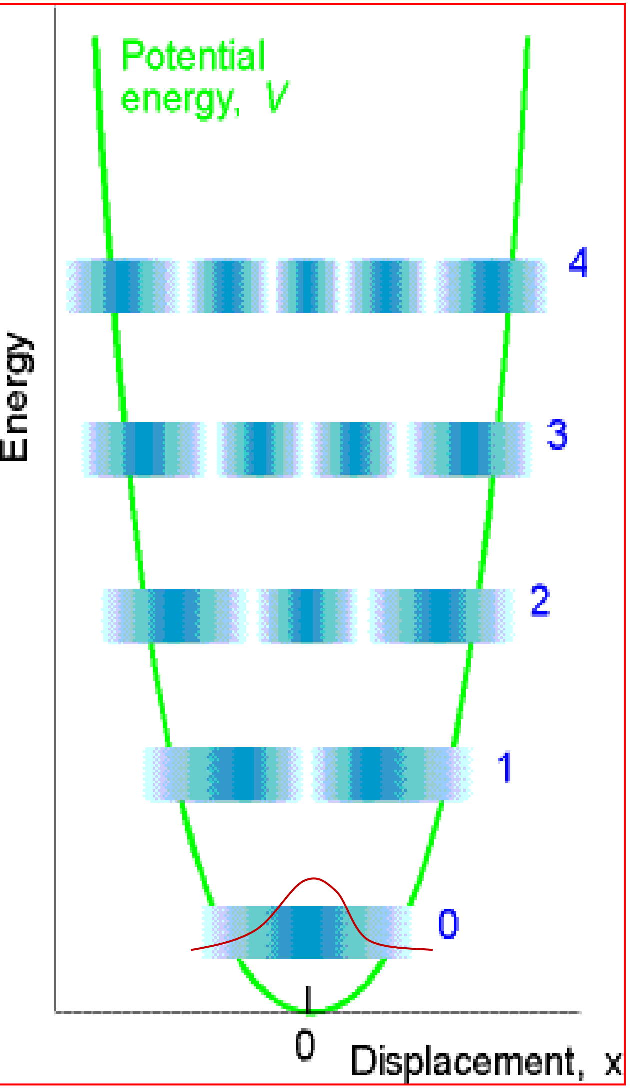
**Spacing** $= h\nu$

# QM harmonic oscillator wavefunction

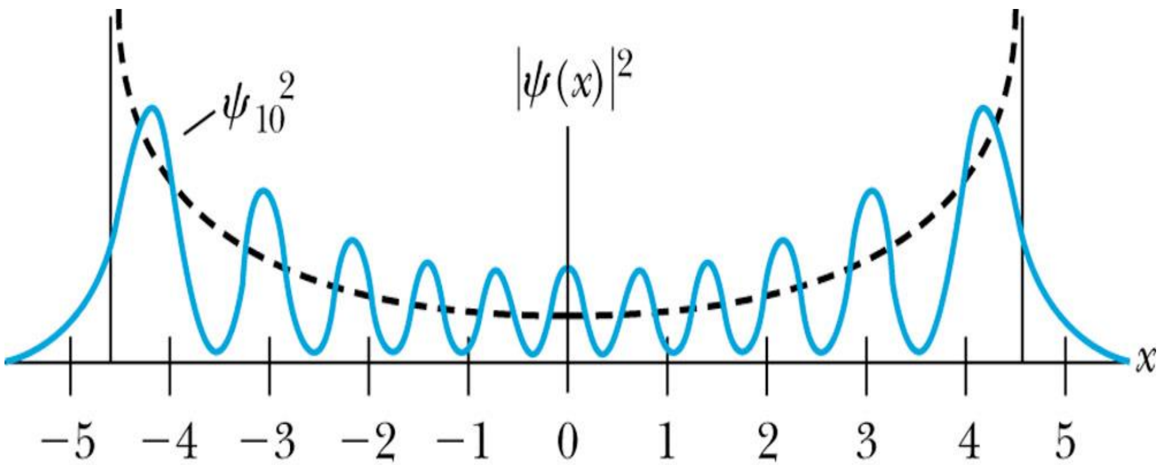
- Number of nodes is n.
- **Wavefunctions alternately symmetric or antisymmetric about x=0**



# QM harmonic oscillator - the probability distributions




- As n increases, probability that the particle is in the centre of the energy well decreases and the probability of its being at the sides of the potential well increases.
- *n* becomes large, regions of highest probability move towards the turning points: Probability distribution resembles the classical result. : Correspondence to classical mechanics.



# QM harmonic oscillator

Determine $\langle E^2 \rangle, \langle E \rangle^2, \langle p_x^2 \rangle, \langle p_x \rangle^2, \langle x^2 \rangle, \langle x \rangle^2$ for the ground state $(\mathrm{n} = 0)$ of a simple harmonic oscillator.